

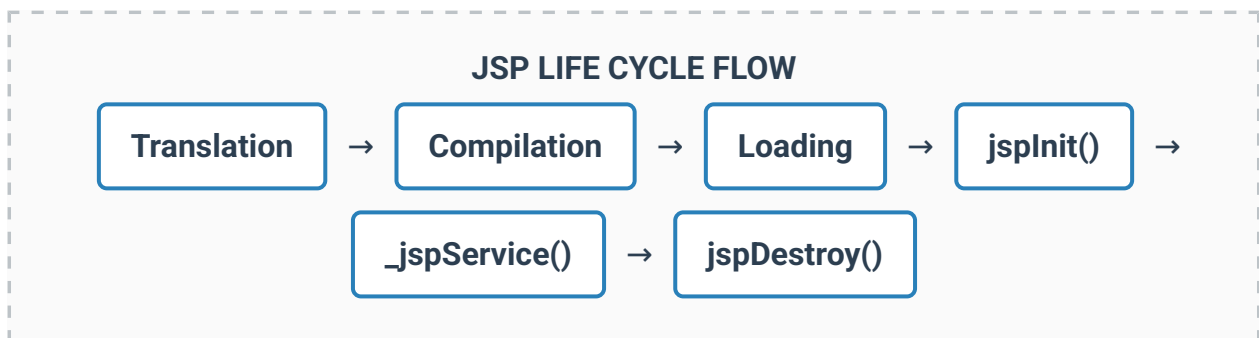
EXAM LAST MIN REVISION UNIT 3

Introduction to JSP

Quick Revision (2 Marks)

JSP is a server-side technology used to create dynamic web pages. It allows HTML and Java code to be written together. A JSP page is converted into a Servlet before execution and is mainly used for the View layer in web applications.

JSP LIFE CYCLE



IMPORTANT EXAM POINTS

- JSP is first converted into a Servlet.
- jsplnit() executes only once.
- _jspService() executes for every request.
- jspDestroy() executes before JSP removal.
- Lifecycle is managed by the web container.
- Developers cannot override _jspService().

Quick Revision (2 Marks)

JSP Lifecycle consists of Translation, Compilation, Loading, Initialization (jsplnit()), Request Processing (_jspService()), and Destruction (jspDestroy()). jsplnit() and jspDestroy() are called once, while _jspService() is called for every client request.

Exam-Ready Version of Your Explanation:

When a user requests a JSP page from the browser, the web container (such as Tomcat) first converts the JSP page into a Servlet source file and then compiles it into a class file. The container loads this class into memory and calls jsplnit() once for initialization. For every client request, the _jspService() method executes and generates a dynamic HTML response using data from databases or business logic. The generated response is sent back to the browser. The Servlet remains in memory and continues serving requests. Finally, when the server shuts down or unloads the JSP, jspDestroy() is called to release resources and perform cleanup.

JSP Syntax & Tags

Quick Revision (2 Marks)

JSP Syntax consists of Directives, Scriptlets, Expressions, and Declarations.

Directives provide instructions to the JSP container and are of three types: Page, Include, and Taglib. They control how a JSP page is processed and compiled.

Before moving to Scriptlet, Expression and Declaration, note that we've already briefly seen them here. In the next topic, we'll study them in much greater detail, including exactly where they are placed in the generated Servlet and the differences between them.

EXAM POINTS

- Scriptlet contains Java statements.
- Expression displays output directly.
- Declaration creates class-level variables and methods.
- Scriptlet and Expression go inside `_jspService()`.
- Declaration goes outside `_jspService()`.
- Expression automatically uses `out.print()`.

Quick Revision (2 Marks)

Scriptlet (`<% %>`) is used to write Java code, Expression (`<%= %>`) is used to display output, and Declaration (`<%! %>`) is used to declare class-level variables and methods. Scriptlets and Expressions are placed inside `_jspService()`, while Declarations become members of the generated Servlet class.

Java Implicit Objects

Most Important Objects for Exams: Focus mainly on request, response, session, out, application.

EXAM POINTS

- Implicit Objects are automatically created by the JSP container.
- No need to create them manually.
- There are 9 JSP implicit objects.
- request receives client data.
- response sends responses.
- session stores user-specific data.
- application stores global data.
- out writes output to browser.

Quick Revision (2 Marks)

JSP Implicit Objects are predefined objects automatically provided by the JSP container. They help in handling requests, responses, sessions, output, configuration, and application-wide data. Common implicit objects include request, response, session, out, and application.

JSP Directives

Page Directive

→ Page settings

Easy Memory Trick

Include Directive

→ Reuse files

Taglib Directive

→ Use custom tags

EXAM POINTS

- Directives use `<%@ ... %>`.
- Processed during JSP translation.
- Do not produce output directly.
- Three directives: Page, Include, Taglib.
- Page Directive controls page settings.
- Include Directive reuses files.
- Taglib Directive enables custom tags/JSTL.

Quick Revision (2 Marks)

JSP Directives are special instructions processed during JSP translation. They are used to control page settings, include external files, and use custom tag libraries. The three JSP directives are Page Directive, Include Directive, and Taglib Directive.

JSP Action Elements & JavaBeans

JSP Action Elements: Built-in JSP tags that help developers perform common tasks without writing a lot of Java code. They are executed during request processing and can dynamically include pages, forward requests, pass parameters, and work with JavaBeans. They make JSP pages cleaner, easier to maintain, and reduce the need for repetitive Java code.

JAVABEANS IN JSP (5 MARKS)

DEFINITION:

JavaBean is a reusable Java class used in JSP to store and manage data. It helps separate business logic from the presentation layer. A JavaBean contains private data members, a public no-argument constructor, and public getter and setter methods.

Example: A Student JavaBean may contain properties like name and age with corresponding getter and setter methods. JSP pages can use this bean to store and display student information.

Conclusion: JavaBeans are used in JSP to encapsulate data and provide a simple, reusable way to transfer information between different components of a web application.

JSP Expression Language (EL)

Easy Memory Trick

```
<%= value %>
```



Old JSP Way

```
${value}
```



Modern EL Way

Think: EL = Easy Lookup (because it easily retrieves data)

ADVANTAGES OF EL

- Less Java code in JSP.
- Easy to read and maintain.
- Accesses request, session, and application data easily.
- Works well with JavaBeans.
- Makes JSP pages cleaner.

EXAM POINTS

- EL stands for Expression Language.
- Syntax: `${expression}`
- Used to access and display data.
- Reduces Java code inside JSP.
- Can access request, session, application, and JavaBeans.
- Supports arithmetic, relational, and logical operators.
- Preferred over Scriptlets for displaying data.

Quick Revision (2 Marks)

JSP Expression Language (EL) is a simple language used to access and display data in JSP pages without writing Java code. It uses the `${}` syntax and can retrieve data from request, session, application scopes, and JavaBeans, making JSP pages cleaner and easier to maintain.